Software Test Plan (STP) — Vehicle Parking System (VPS)

Project	Vehicle Parking System (VPS)
Version	**1.1** (updated after Sprints 1–2; supersedes v1.0 of 16-09-2026)
Authors	QA Team — Himansh Ahuja (QA Lead), Angelo Arakal, Shweta Mailaragouda Patil, Akarsh Parashuram
Date	02-11-2026
Status	Approved for Sprint 3 test execution

Revision history

Version	Date	Author	Change summary
1.0	16-09-2026	QA Team	Initial test plan (Part 1)
1.1	02-11-2026	Himansh Ahuja	Test cases updated after the sprints (38 cases, was 16); tools aligned with the implementation; schedule moved to the sprint calendar; traceability regenerated from automated test tags; performance environment re-specified

1. Introduction

1.1 Purpose

This plan defines the objectives, scope, strategy, environment, schedule and responsibilities for verifying VPS v1.1 against SRS v1.0 and SAD v1.0, including functionality, performance, security and resilience.

1.2 Scope

Testing covers vehicle entry/exit automation (ANPR, signed QR tickets, barrier and ground-loop safety), slot allocation and sensor-driven signage, dynamic fares (tiered rates, grace period, surge, coupons, GST), cashless payments and receipts, 48-hour reservations with no-show release, the admin console, reports, RBAC/MFA, audit logging and 72-hour edge offline operation.

Excluded: core banking switch clearing, physical enclosure durability, towing/clamping enforcement, and physical hardware (replaced by simulators — see `docs/Deviations.md` DEV-02).

1.3 References

- VPS Software Requirements Specification v1.0 (09-09-2026)
- VPS Software Architecture and Design Specification v1.0 (09-09-2026)
- `docs/Deviations.md` — deviations recorded during the sprints
- `docs/testing/Test_Case_Specification.md` — detailed test cases (generated, v1.1)

1.4 Definitions

VPS, ANPR, VMS, RTM, POS, TOTP (time-based one-time password), WAL (SQLite write-ahead log), p90/p95 (90th/95th percentile latency), SUT (system under test).

2. Test items

Item	Source	Owner
Ingress module, lane API, entry kiosk	`vps/ingress.py`, `vps/api/ingress.py`, `web/entry.html`	Angelo
Security core: signer, PII vault, Argon2id, TOTP, TLS runner	`vps/security.py`, `vps/tls.py`	Angelo
Authentication, RBAC, audit log	`vps/auth.py`, `vps/audit.py`, `vps/api/auth.py`	Angelo
Hardware simulators, slot allocation, sensor hub & VMS, alerts	`vps/hardware.py`, `vps/allocation.py`, `vps/sensors.py`, `vps/alerts.py`	Shweta
Edge store-and-forward controller	`vps/edge_queue.py`, `vps/api/edge.py`	Shweta
Tariff engine, payment gateway adapter, egress, receipts	`vps/tariff.py`, `vps/payment.py`, `vps/egress.py`, `vps/receipt.py`	Akarsh
Reservations, scheduler, admin console, reports	`vps/reservation.py`, `vps/scheduler.py`, `vps/api/admin.py`, `vps/reports.py`	Himansh

3. Features to be tested

Requirement	Feature	Test cases
VPS-F-001	ANPR capture ≤ 1.5 s, normalisation, unreadable-plate path	TC-ENT-01, TC-ENT-06, TC-SYS-04
VPS-F-002	Signed QR ticket ≤ 2 s, unique IDs, anti-passback	TC-ENT-02, TC-ENT-05
VPS-F-003 / F-004	Barrier ≤ 500 ms; loop safety halt and auto-lower	TC-ENT-03, TC-ENT-04
VPS-F-005 – F-007	Nearest-slot allocation, lot full, concurrency; LED ≤ 2 s; VMS ≤ 1 s	TC-SLOT-01 … 04
VPS-F-008 – F-010	Tiered fares and caps; 15-min grace; surge and coupons	TC-BILL-01 … 03, TC-SYS-03
VPS-F-011 – F-014	Exit lookup ≤ 1 s; UPI/card/wallet; barrier on PAID only; GST receipts	TC-PAY-01 … 05, TC-SYS-01
VPS-F-015 / F-016	48 h reservations, check-in rules, 30-min no-show release	TC-RES-01 … 03, TC-SYS-02
VPS-F-017 / F-018	Admin configuration without restart; exports ≤ 5 s	TC-ADM-01, TC-ADM-02, TC-SYS-05
VPS-NF-001	Ingress p90 ≤ 3 s	TC-Perf-01
VPS-NF-002	Availability (lab proxy: soak test)	TC-Rel-01
VPS-NF-003	10,000 slots, 50 lanes, p95 < 200 ms at 500 req/s	TC-Scale-01
VPS-NF-004	No plaintext card data or credentials	TC-SEC-06
VPS-NF-005	72 h offline queuing, ordered idempotent sync ≤ 5 min	TC-Fault-01
VPS-SR-001 … 005	TLS 1.3, Argon2id/AES-256, RBAC+MFA, HMAC QR, immutable audit	TC-SEC-01 … 05

4. Features not to be tested

- Core banking clearing (gateway partner responsibility); real Razorpay/Stripe settlement.
- Physical durability, weatherproofing, towing and clamping.
- RFID pass-holder opening (not implemented — DEV-07) and multilingual audio prompts (DEV-09).
- Real monthly uptime (production monitoring only — verified partially via TC-Rel-01, DEV-10).

5. Test approach / strategy

5.1 Levels

Level	What	How	Location
Unit	Each module in isolation: fare rules, HMAC signing, allocation, queue semantics, RBAC, etc.	pytest, in-memory SQLite, fake clock, hardware simulators	`tests/unit`
Integration	API ↔ services ↔ DB; signed payment webhooks; lane hardware via API	FastAPI TestClient	`tests/unit/test_*_api.py` , API classes
System	Complete driver / attendant / admin journeys; security validation against a live TLS server; resilience	TestClient + live uvicorn TLS 1.3 server	`tests/system`
Performance	NF-001 latency, NF-003 load at 10k slots / 50 lanes, F-018 export volume	Separate server process + httpx thread pool	`tests/perf`
Acceptance (UAT)	Facility-manager and attendant walkthrough of the six web screens against the SRS acceptance criteria	Manual, recorded in the sprint demo videos	`demo/`

5.2 Types

Functional, boundary-value (grace period, 48 h window, no-show cut-off, surge window, daily cap), negative (invalid plates, counterfeit tickets, bad signatures), concurrency (90 parallel allocations), regression (full suite on every push via GitHub Actions), performance/load, resilience/fault-injection (WAN loss, lane controller loss, gateway outage), and security.

5.3 Entry and exit criteria

- **Entry:** story merged to `main` ; CI green; test data and simulators available.
- **Exit:** 100 % of test cases executed; zero open critical/high defects; all security and performance NFR cases pass; statement and branch coverage ≥ 90 % (enforced in CI).

5.4 Security validation

Requirement	Validation
VPS-SR-001	Live server started with the TLS 1.3-only context; TLS 1.3 client succeeds, TLS 1.2 client refused, plain HTTP not served, HSTS present; 426 when TLS is required behind a proxy
VPS-SR-002	Inspect stored hashes (`$argon2id$`) and ciphertexts; full database dump searched for plaintext plates, phones, card numbers, passwords and TOTP secrets

Requirement	Validation
VPS-SR-003	RBAC matrix (anonymous, attendant, manager, admin × protected endpoints); TOTP mandatory for admin/manager; lockout after 5 failures; session expiry
VPS-SR-004	Tampered QR body, forged signature and re-signed clone rejected with a CRITICAL alert and an audit entry
VPS-SR-005	Database refuses UPDATE/DELETE on the audit table; out-of-band edits detected by hash-chain verification; retention ≥ 3 years

6. Test environment

Category	Items
Hardware	Developer laptops and GitHub-hosted Ubuntu runners; gate hardware simulated (`vps/hardware.py`)
Software	Python 3.11/3.12, FastAPI, uvicorn (uvloop), SQLite 3 (WAL), Chromium (Playwright) for UI recording
Tools	pytest, pytest-cov (branch), pytest-html, httpx, ruff, GitHub Actions, GitHub Issues/Projects
Test data	Indian plate formats (incl. BH series), seeded tariffs/surge/coupons, test cards 4111…1111 / 5555…4444, sandbox webhooks signed with the test secret, 10,000-bay layout for load tests

7. Test schedule

Milestone	Date
Test cases updated for sprint changes (this plan + specification v1.1)	02-Nov-2026
Unit and integration execution (continuous in CI since Sprint 1)	19-Oct → 06-Nov-2026
System, security and resilience execution	03-Nov-2026
Performance and scalability execution	04-Nov-2026
Reports generated into `docs/reports/`	05-Nov-2026
Final test sign-off	06-Nov-2026

8. Test deliverables

- This plan (v1.1) and the Test Case Specification v1.1
- Automated suites: `tests/unit` , `tests/system` , `tests/perf`
- **Unit Test Report**, **System Validation Report**, **Code Coverage Report**, **Branch Coverage Report** — `docs/reports/`
- HTML test reports and line-level coverage — `docs/reports/html/`
- Deviation log — `docs/Deviations.md`

9. Roles and responsibilities

Role	Name	Responsibility
QA Lead	Himansh Ahuja	Test plan, test case specification, performance suite, report generation, sign-off coordination
Test Engineer	Angelo Arakal	Security validation suite; unit tests for security, auth, audit and ingress

Role	Name	Responsibility
Test Engineer	Shweta Mailaragouda Patil	Resilience suite (edge offline, controller and gateway outage, soak); unit tests for hardware, allocation, sensors and edge
Software Engineer	Akarsh Parashuram	End-to-end journey suite; unit tests for tariff, payment, egress and receipts; defect fixes
Evaluator	Prof. Pavithra	Review outcomes and approve sign-off readiness

10. Risks and mitigation

Risk	Mitigation
Physical gate hardware unavailable	Interface-compatible simulators; UI and API controls to inject loop, sensor and fault events
Flaky timing on shared CI runners	Business-rule tests use a fake clock; performance job publishes metrics but does not block merges
Payment gateway sandbox unavailable	In-repo sandbox with the same signed-webhook contract
Load generator skewing server measurements	Server runs as a separate process; warm-up phase excluded from measurement

11. Assumptions and dependencies

- Course calendar: Sprint 1 19–23 Oct, Sprint 2 26–30 Oct, testing 2–6 Nov 2026.
- Production deployment would replace SQLite with PostgreSQL and the sandbox with Razorpay/Stripe (DEV-01, DEV-04).

12. Suspension and resumption criteria

- **Suspend** if CI is red on `main` for more than 4 hours, or blocking defects affect more than 30 % of ingress/egress cases.
- **Resume** once the blocking defect fix is merged and the smoke subset (`pytest tests/system -k walk_in`) passes.

13. Test case management and traceability

Test cases are maintained in `scripts/test_catalog.py` and rendered into the Test Case Specification. Every automated test carries `@pytest.mark.tc("TC-…")` and `@pytest.mark.req("VPS-…")` markers. The report generator reads these from the JUnit output to build the updated RTM in the System Validation Report.

Area	Test cases	Count
Ingress	TC-ENT-01 … 06	6
Slots & signage	TC-SLOT-01 … 04	4
Billing	TC-BILL-01 … 03	3
Payment & exit	TC-PAY-01 … 05	5

Area	Test cases	Count
Reservations	TC-RES-01 … 03	3
Admin & reports	TC-ADM-01, 02	2
Non-functional	TC-Perf-01, TC-Rel-01, TC-Scale-01, TC-SEC-06, TC-Fault-01	5
Security	TC-SEC-01 … 05	5
System journeys	TC-SYS-01 … 05	5
Total		**38**

14. Test metrics and reporting

- Executed / passed / failed / skipped per suite and per test file.
- Test case pass rate and requirement verification status (verified / partial / failed).
- Statement and branch coverage per module and per team member.
- Latency percentiles (p50/p90/p95/p99), throughput and error count for performance runs.
- Defects found and fixed during testing.

15. Approvals

QA Lead (Himansh Ahuja): _____

Course Coordinator (Prof. Sheela Devi): _____

Project Evaluator / Guide (Prof. Pavithra): _____